American Field SDR

A crowdsourcing white paper for a safe, fieldable RF research appliance

Prepared for discussion by **C. Williams / 0m364**

Scope. This paper proposes a practical architecture for a portable SDR platform that is functional, auditable, and realistic about hardware limits. It recommends an FPGA-centered transceiver design over a pocket-gadget design, and it keeps transmission tightly constrained by policy and hardware.

What this document is. A sanitized architecture note with board partitioning, interconnect diagrams, power domains, and prototype paths suitable for crowdsourced review.

What this document is not. It is not a manufacturing package, not a pin-level wiring diagram, and not a how-to for offensive or unlawful RF use.

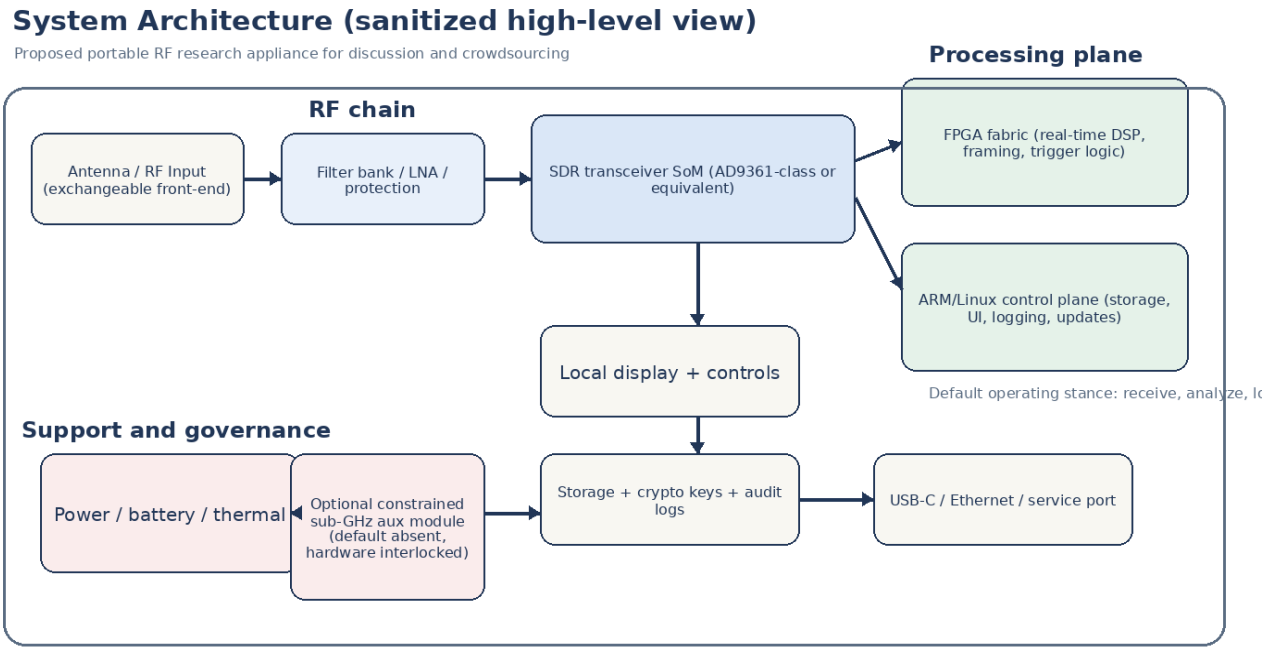

Figure 1. Proposed system architecture for a lawful, receive-first field SDR platform.

Version: April 2026

1. Executive summary

The recommended direction is **not** a literal clone of a consumer pocket tool. It is a compact field instrument centered on a real SDR transceiver system-on-module, FPGA fabric for deterministic signal handling, and an ARM/Linux control plane for storage, logging, UI, and updates. This shifts the design from "gadget" thinking toward "instrument" thinking: stronger receive capability, cleaner processing boundaries, and better governance.

For a serious prototype, the strongest reference class is an **AD9361 plus Zynq** module rather than an RX-only USB receiver. Airspy R2 remains a useful benchmark for receive-side expectations, but its official specifications make clear that it is a **receiver** with 24 to 1700 MHz native RX coverage and 10 MSPS IQ output, not an integrated transceiver core.[1] By contrast, Analog Devices' ADRV9361-Z7035 combines an AD9361 RF Agile Transceiver with a Xilinx Zynq-7000 SoC and exposes 2x2 receive and transmit paths from 70 MHz to 6.0 GHz, which is much closer to the target architecture.[2]

This paper recommends three staged build paths: (a) a low-cost proof platform based on PlutoSDR-class hardware for algorithm and UI work, (b) a serious prototype based on an AD9361-class transceiver SoM with Zynq, and (c) a fieldable appliance that takes cues from Epiq's compact Matchstiq line, where local signal processing happens on the SoC rather than on a tethered laptop.[3][4]

To keep the design suitable for open discussion, this document deliberately omits pin maps, RF switch truth tables, matching-network values, power-amplifier details, and other build-ready transmission specifics. The result is detailed enough for contributors to critique architecture, packaging, thermal strategy, power domains, and software boundaries without turning into a manufacturing recipe.

Design stance: receive first, process locally, log everything, and make any optional transmit capability physically separable, power-gated, and policy-constrained.

2. Goals, non-goals, and document boundary

Category	Included in this white paper	Explicitly omitted
Product intent	Portable RF research instrument; field logging; offline analysis; modular development path	Pocket-ready consumer product; mass-market "magic wand" clai
Hardware detail	High-level architecture; board partitions; safe interconnect classes; power domains; RF blocks	Pin maps; power-rail networks; PCB stackups; schematic valu
Operational stance	Receive-first modes; audit logging; constrained auxiliary modes; local processing	Ready-to-use transmit firmware; wideband arbitrary transmis
Crowdsourcing target	Mechanical design, UI, FPGA pipeline ideas, power strategy, test plans, capture formats	Jurisdiction-specific compliance claims; evasion

A useful crowdsourcing paper needs enough specificity to invite engineering critique, but not so much that it becomes a copy-and-build shortcut. This is why the diagrams below are intentionally "schematic-style" rather than manufacturing-ready.

3. Reference platforms that informed the architecture

Platform	Official positioning / core facts	Why it matters here
Airspy R2	RX-only SDR receiver; 24 to 1700 MHz native RX coverage; 10 MSPS IQ output; external 2.5 MSPS mode for low-power	Useful benchmark for receive-side expectations, but not the right architecture for a serious prototype.
HackRF One	Open source half-duplex transceiver; 1 MHz to 6 GHz; up to 20 MSPS half-duplex IQ[5]	Good for comparison on bandwidth flexibility, but less aligned with deterministic
ADALM-PLUTO	AD9363 plus Zynq-7010; 325 MHz to 3.8 GHz; up to 56 MHz instantaneous bandwidth; 1 TX and 1 RX[6]	Best low-cost proof platform for DSP, UI, and workflow experiments.
ADRV9361-Z7035	AD9361 plus Zynq-7000; 2x2 RX/TX; 70 MHz to 6.0 GHz[2]	Closer off-the-shelf reference for a serious prototype.

Platform	Official positioning / core facts	Why it matters here
Matchstiq Z3u	Compact field-ready RF transceiver and signal processing platform; Zynq UltraScale+ SoC [4]	Reference for the appliance class: rugged, local, and portable rather

The table above explains the central architectural move in this paper: **use receiver-class products as performance references, but use transceiver-plus-FPGA products as the design pattern**. That move is what takes the concept from "better toy" to "credible instrument."

4. Proposed architecture

The recommended core is an **SDR transceiver SoM** feeding **FPGA fabric** for real-time signal handling and an **ARM/Linux plane** for orchestration. In a Zynq UltraScale+ direction, AMD's documentation highlights secure-boot and security hardware in the Configuration Security Unit, including secure boot, tamper monitoring and response, secure key storage and management, and crypto acceleration.[7][8] Those features matter more to a field appliance than cosmetic miniaturization.

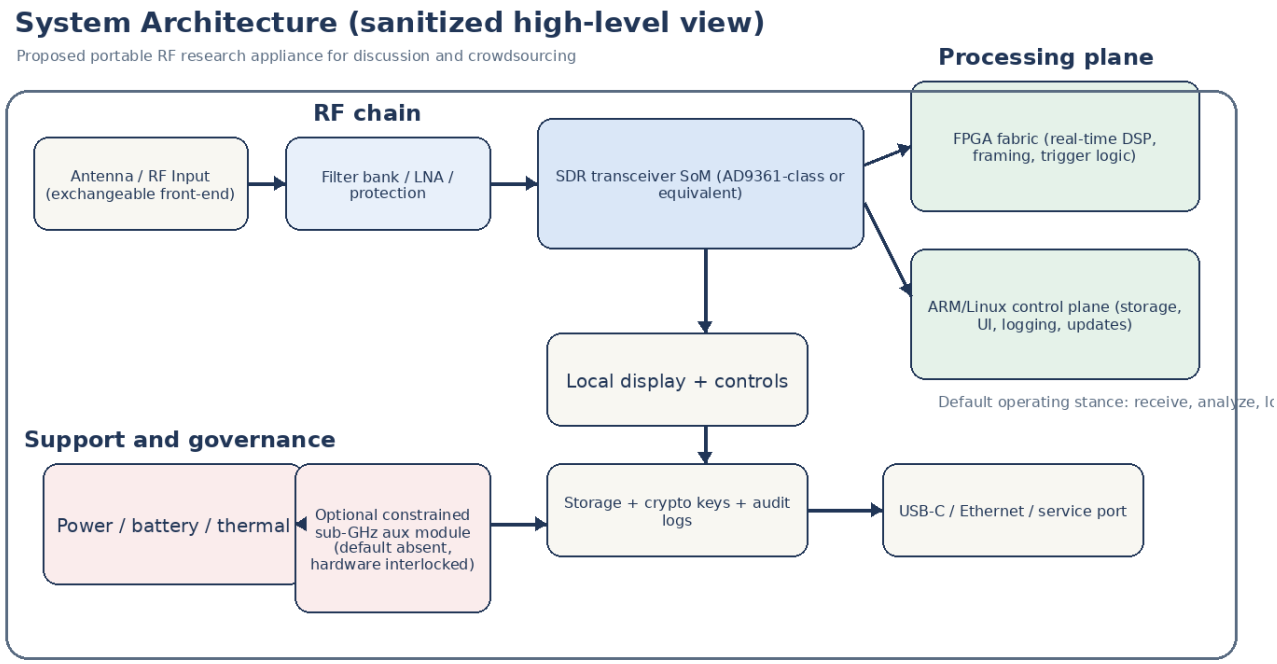

Figure 2. High-level architecture. The auxiliary module is optional and absent in the default receive-focused assembly.

Recommended control-plane assumptions

- Local-first workflow: capture, classify, store, and export without requiring cloud services.
- Read-only or tightly managed root filesystem for field operation, with signed update bundles and rollback images.
- Audit trail for mode changes, removable media use, and any activation of constrained auxiliary hardware.
- Operator-visible state: receive, maintenance, and bench-lab modes should be impossible to confuse at a glance.

5. Board partitioning and safe interconnect view

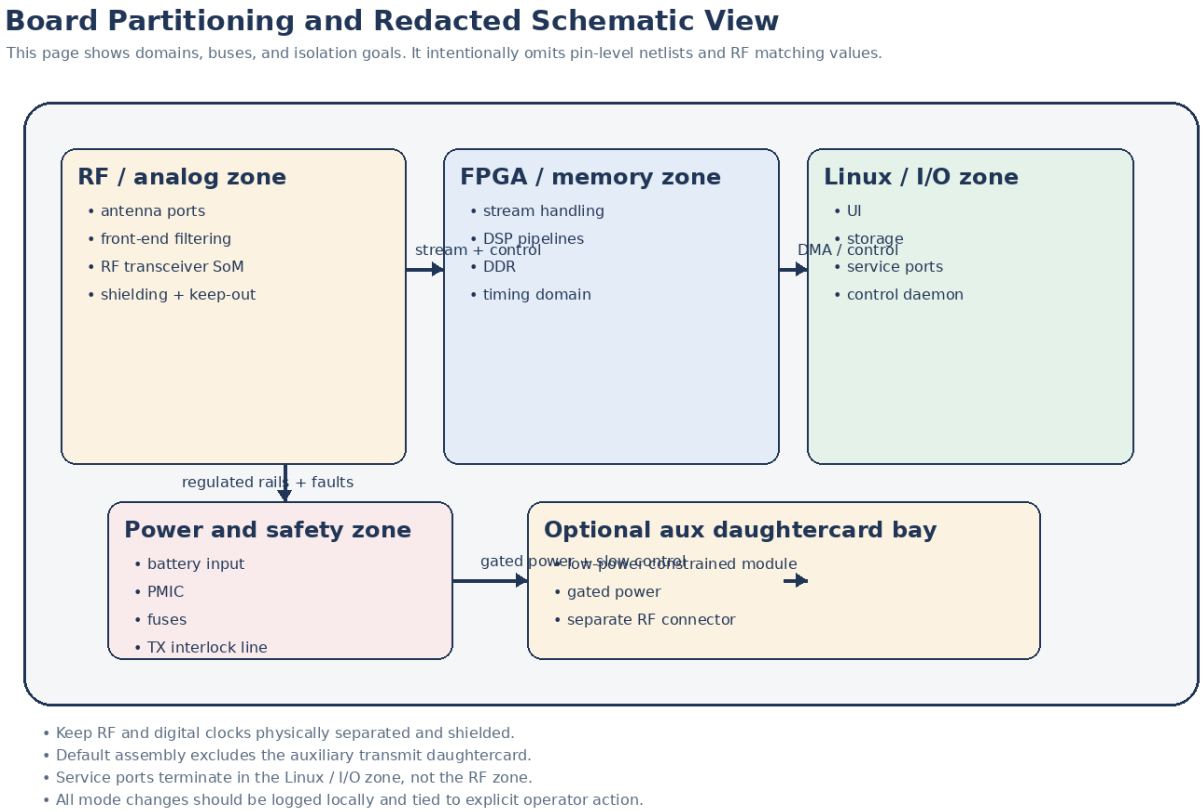

Figure 3. Board zones, isolation goals, and domain boundaries. This is intentionally not a pin-level schematic.

The board should be treated as four major domains: RF/analog, FPGA/memory, Linux/I/O, and power/safety. The most common amateur mistake is to let software convenience dictate physical placement. A better approach is to force the layout to respect clock, shielding, and heat boundaries first, then let the software team adapt. Humans hate that because software people always think the board is a rumor until EMI proves otherwise.

High-level interconnect classes

Link	Purpose	Safe level of detail for crowdsourcing
RF front end <-> SDR SoM	Tuned RF path and front-end selection	Connector class, isolation goal, and need for shielding. No matching values
SDR SoM <-> FPGA fabric	Sample stream plus low-level control	Parallel/high-speed data path and control relationship. No lane or timing det
FPGA fabric <-> ARM/Linux	DMA, telemetry, configuration, and logging	Functional split between deterministic DSP and operating-system services.
Power/safety <-> aux bay	Gated rails and hard interlock	Power-domain concept, fault handling, and default-disabled state.

6. Interconnect, deployment paths, and power strategy

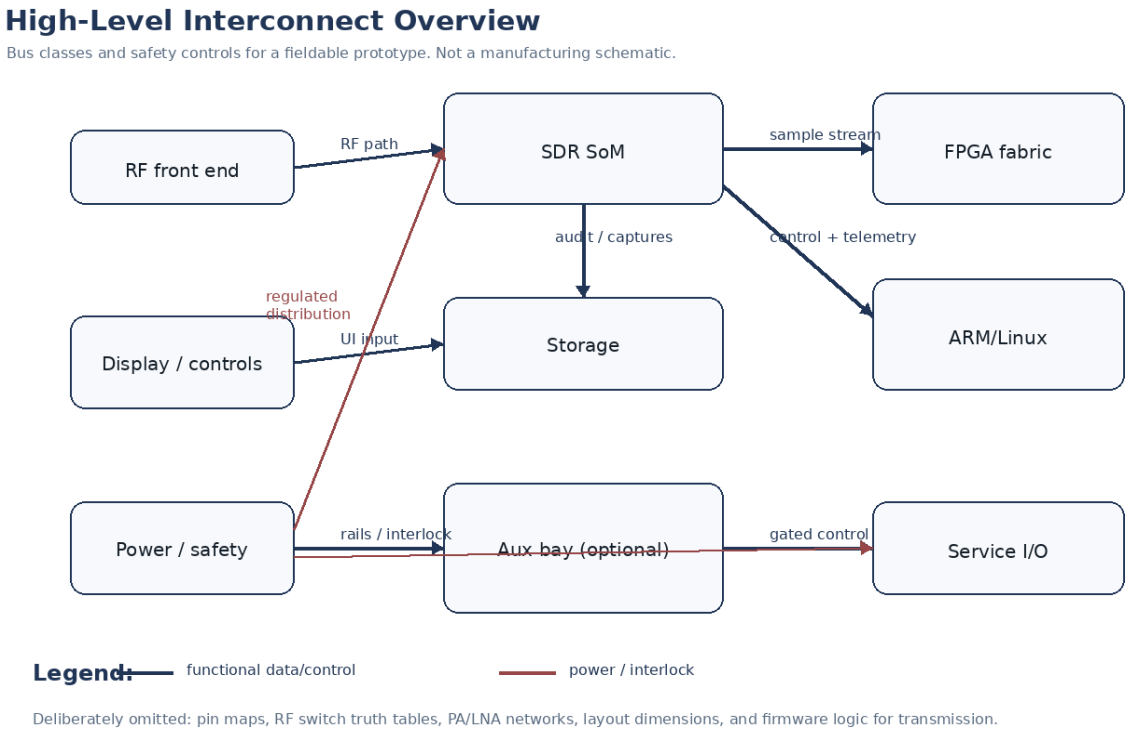

Figure 4. Functional interconnect overview with data/control paths and power/interlock paths shown separately.

Power design should assume three realities: SDR work is bursty, Linux workloads are messy, and portable enclosures trap heat. That means the project needs explicit power domains, current limiting, and thermal telemetry from the first prototype rather than as a late-stage patch. Crowdsourcing is most valuable here for mechanical design, airflow, thermal spreader concepts, and serviceability.

Staged deployment options

Tier	Reference hardware path	Best use
A. Proof platform	ADALM-PLUTO-class or other AD9363/Zynq dev board.[6]	DSP experiments, UI flow, logging pipeline, basic enclosure concepts.
B. Serious prototype	ADRV9361-Z7035-class SoM or equivalent.[2]	Integrated RF, FPGA, and Linux bring-up in a realistic architecture.
C. Fieldable appliance	Custom carrier around transceiver SoM, with packaging, ventilation, battery systems, and [illegible]	Ruggedization, battery systems, thermal discipline, service workflow, sec

Arch Linux remains viable as a lab-facing environment and even as a prototype convenience on supported boards, but a production Zynq device normally lives more comfortably in a vendor-supported embedded Linux flow. Arch Linux ARM's ZedBoard support shows that Arch on Xilinx-class hardware is possible for prototyping, while Arch Linux ARM's own downloads page also reminds us that it is a rolling distribution, which is wonderful for a bench and less wonderful for a field appliance unless you wrap it in strong release discipline.[9][10]

7. Security and governance controls

- Default to receive-focused firmware and assembly; optional auxiliary module absent by default.
- Separate service-mode and field-mode images or profiles, with obvious visual indicators.

- Use secure boot or measured boot where the chosen SoC supports it; anchor local audit logs to signed images when feasible.[7][8]
- Treat removable storage, service ports, and update media as controlled interfaces rather than convenience features.
- Keep remote administration disabled by default. Service access should be deliberate, local, and logged.

None of that is glamorous, which is exactly why it matters. The best field tools are often boring in all the right ways: predictable states, clear logs, no silent mode drift, and no magical hidden features that turn into legal or operational surprises later.

8. Contribution targets for an open review

- Carrier-board partitioning review and connector strategy
- Battery and thermal architecture for 2 to 4 hour field sessions
- UI information hierarchy for portable spectrum review and capture management
- FPGA pipeline partition suggestions for trigger logic, framing, and pre-processing
- Mechanical design concepts for a slab or bag-node enclosure
- Test-plan proposals for power, EMI, logging integrity, and serviceability

This is where a crowd can help without derailing the project. Ask contributors to improve packaging, safety, reliability, workflow, and operator clarity. Do not ask the internet to "make it do everything," because the internet reliably answers that request by stuffing a product with irresponsible edge cases and calling it innovation.

References

[1] Airspy, “Airspy R2,” official product page. Native RX coverage 24 to 1700 MHz and 10 MSPS IQ output; low-power 2.5 MSPS mode is marked experimental.

[2] Analog Devices, “ADRV9361-Z7035 Evaluation Board,” official product page. AD9361 plus Zynq-7000; 2x2 receive and transmit paths from 70 MHz to 6.0 GHz.

[3] Analog Devices, “ADALM-PLUTO Evaluation Board,” official product page. AD9363 plus Zynq-7010; 325 MHz to 3.8 GHz; up to 20 MHz instantaneous bandwidth; 1 TX / 1 RX.

[4] Epiq Solutions, “Matchstiq Z3u” and “Matchstiq Z Series,” official product pages. Compact field-ready RF transceiver plus signal processing on Zynq-based SoC platforms.

[5] Great Scott Gadgets, “HackRF One,” official product page. Half-duplex transceiver from 1 MHz to 6 GHz; up to 20 MSPS; 8-bit I/Q.

[6] Analog Devices, “AD9361 Datasheet and Product Info,” official product page. 2x2 transceiver; TX 47 MHz to 6.0 GHz; RX 70 MHz to 6.0 GHz; tunable channel bandwidth below 200 kHz to 56 MHz.

[7] AMD, “Configuration Security Unit (CSU),” Zynq UltraScale+ MPSoC Software Developer Guide. Secure boot, tamper monitoring/response, secure key storage/management, and crypto acceleration.

[8] AMD, “Secure Boot,” Embedded Design Tutorial for Zynq UltraScale+ MPSoC. Hardware root of trust and partition encryption/authentication flow.

[9] Arch Linux ARM, “ZedBoard,” official platform page. Demonstrates Arch Linux ARM support on a Xilinx Zynq-7000 platform for prototyping.

[10] Arch Linux ARM, “Downloads,” official page. Notes that the distribution is rolling and updated via pacman -Syu.

Labeling note: This document is intentionally labeled **C. Williams / 0m364** throughout for discussion circulation. It is a white paper for critique, not a release package or manufacturing file set.